

FarmAssist

Sophie • your digital agriculture officer

An algorithm-based agricultural knowledge search system

Team 20

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Course project for DSA-3 (25CS2103E). The system answers farming questions about crops, diseases, pests and fertilizers. There is no machine learning and no network call anywhere in it: every answer is produced by eight classical algorithms running over a local agricultural database.

1. Overview

FarmAssist is a console assistant, named Sophie, that a farmer talks to in plain language. A question such as *"my tomatoe has yellow leaves"* is corrected for spelling, expanded through a table of local crop names, scanned for every crop, disease, pest, symptom and fertilizer it mentions, routed to whichever feature can answer it, and answered from the local database.

The point of the project is that the intelligence is algorithmic, not statistical. Nothing is delegated to the Java library: there is no `String.contains()` and no `Collections.sort()`. String search, spelling repair, ranking, budgeting and assignment are each implemented from first principles, and the program prints a live trace of which algorithm ran at every step so the usage is visible while it is being demonstrated.

2. The system at a glance

Knowledge base	Size	Derived index	Size
Crops	63	Search index	271 documents
Diseases	64	Spelling dictionary	974 words
Pests	44	Pattern trie	285 names
Fertilizers	38	Local name synonyms	396
Articles	62	Symptom keywords	76

The search index is larger than the article count because a profile document is generated for every crop, disease, pest and fertilizer record and indexed alongside the 62 hand written articles.

Package	Lines of Java	Responsibility
algo	691	The eight algorithms, pure and free of project logic
engine	1530	Loading, spelling, entity extraction, routing, search, diagnosis, planning
ui	1735	The chat loop and every screen it draws
model	126	Crop, Disease, Pest, Fertilizer, Article records
util	98	The algorithm trace
Total	4180	30 source files, no libraries and no build tool

3. Where each algorithm is used

Every one of the eight is load bearing. None of them is present only to satisfy the brief; removing any one of them removes a feature the farmer can see.

#	Algorithm	Where it does the work
1	KMP	Exact matching of a symptom or damage keyword inside a disease or pest record, and locating the precise position of a term inside a document so the snippet can be cut around it
2	Rabin-Karp	Rolling hash scan across all 271 indexed documents, counting how often each term occurs; the counts become the relevance score

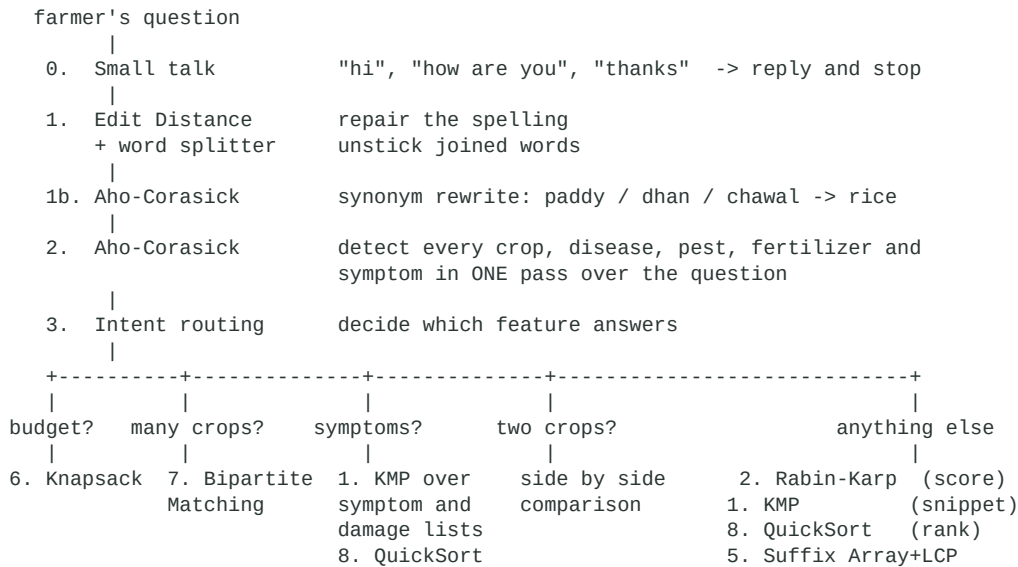
#	Algorithm	Where it does the work
3	Aho-Corasick	One single pass over the question detects every crop, disease, pest, fertilizer and symptom at once against 285 patterns; a second automaton rewrites local names, so paddy becomes rice and bhindi becomes okra
4	Edit Distance	Spelling repair (tomatoe to tomato), tolerating typos in greetings, and the "closest thing I have" fallback when a search finds nothing
5	Suffix Array + LCP	Related reading: article similarity measured as the longest common substring between two article bodies, built with Kasai's algorithm
6	0/1 Knapsack	The best basket of fertilizer inside a stated budget, where price is the weight and the benefit score is the value
7	Bipartite Matching	Giving each of several crops a different fertilizer, by Kuhn's augmenting path method
8	Randomized QuickSort	Ranking search results, candidate diseases, candidate pests and related articles

A ninth helper sits beside them: a word segmentation dynamic program that splits joined words, turning lateblight into late blight. It is deliberately not counted among the eight, because Edit Distance genuinely cannot repair a missing space.

4. Complexity

Algorithm	Time	Space
KMP	$O(n + m)$	$O(m)$
Rabin-Karp	$O(n + m)$ expected	$O(1)$
Aho-Corasick	$O(n + M + z)$, M = total pattern length, z = matches	$O(M)$
Edit Distance	$O(n \times m)$ per word pair	$O(n \times m)$
Suffix Array	$O(n \log^2 n)$ to build, $O(n)$ for the LCP by Kasai	$O(n)$
0/1 Knapsack	$O(\text{items} \times \text{budget})$	$O(\text{items} \times \text{budget})$
Bipartite Matching	$O(V \times E)$	$O(V + E)$
Randomized QuickSort	$O(n \log n)$ expected	$O(\log n)$ stack

5. The pipeline every question travels



Routing is the only place where the features meet. Each branch below it is independent, which is why a new feature costs one intent and one screen.

6. What Sophie can answer

Intent	Example question	Driven by
CROP_INFO	how to grow rice	Record lookup + article search
DISEASE_DIAGNOSIS	my tomato has yellow leaves	KMP, QuickSort
FERTILIZER_BUDGET	suggest fertilizer for tomato under 3000	0/1 Knapsack
CROP_FERTILIZER_MATCH	match fertilizers for rice cotton banana	Bipartite Matching
CROP_PLANNER	which crop for low rainfall	Rainfall, temperature and water filter
CROP_COMPARE	compare rice and wheat	Side by side record table
PEST_INFO	stem borer	Record lookup
FERTILIZER_INFO	what is dap	Record lookup
GENERAL_SEARCH	anything else	Rabin-Karp, KMP, QuickSort, Suffix Array

7. The terminal interface

The whole screen is drawn through a single class, `ui/Theme.java`, which owns one 78 column grid. Panels, tables, headings, cards and rules are all measured against that grid and share one left margin, so no screen can drift out of alignment with another. Padding is measured in visible columns rather than bytes, which is what keeps a coloured cell from collapsing its column.

Colour is 24 bit, with an automatic fall back to the classic 16 and to plain text; every box drawing character has an ASCII twin. Both are detected at start up and can be overridden from the chat with `color basic`, `color off` or `ascii on`. The transcripts below are the program's own ASCII and no-colour output, so they are exactly what a plain console shows.

```
+-----+
|                                     |
|               S O P H I E         |
|                                     |
|   FarmAssist . your digital agriculture officer   |
|                                     |
+-----+

loading the knowledge base ...

| KNOWLEDGE BASE -----
*  Crops           63           search index      271 documents
!  Diseases        64           spell dictionary  974 words
%  Pests           44           pattern trie      285 names
#  Fertilizers     38           local names       396 synonyms
-  Articles        62           crop facts        63 profiles
```

Start up: the masthead and what was loaded into memory.

```
* Sophie | The best basket for tomato within Rs 3000:

+- FERTILIZER PLAN -----+
| FERTILIZER                NPK                COST      BENEFIT |
+-----+
| NPK 10-26-26              10-26-26          Rs 1200      80/100 |
| Farmyard Manure           1-1-1             Rs 250        60/100 |
| Vermicompost              1-1-1             Rs 400        65/100 |
| Micronutrient Mixture     0-0-0             Rs 500        48/100 |
| Liquid Biofertilizer      2-1-1             Rs 350        54/100 |
| Phosphate Solubilising Bacteria 0-1-0          Rs 170        53/100 |
| TOTAL                     Rs 2870           366 |
+-----+
```

0/1 Knapsack choosing the best basket of fertilizer inside a budget.

```
* Sophie | Walking your field with yellow, brown spot, spot on tomato ...

| WHAT IS LIKELY WRONG -----
1  EARLY BLIGHT                ##### score 18
   attacks    tomato, potato -> includes your crop
   symptoms    concentric rings on leaves, brown spots, yellowing around
                the spots, leaf drop, target board pattern
   matched     yellow, brown spot, spot
   treatment   Spray mancozeb 0.25 percent every ten days and remove the
                lower infected leaves.

2  BROWN SPOT                  #####.. score 14
   attacks     rice
   symptoms    brown oval spots, dark brown lesions, weak grain filling
   matched     spot
   treatment   Apply mancozeb 0.25 percent. Correct potassium and zinc
                deficiency in the soil.
```

Diagnosis: candidates ranked by QuickSort, each scored by KMP matches.

8. Data files

Every file uses the vertical bar as its separator and a leading hash for comments. Rows added to any of them are picked up on the next run with no code change.

File	Rows	Contents
crops.csv	63	Season, soil, water, N-P-K, duration, temperature, rainfall, spacing, varieties, yield
diseases.csv	64	Affected crops, symptom phrases, treatment

File	Rows	Contents
pests.csv	44	Affected crops, damage phrases, control
fertilizers.csv	38	N-P-K, price per bag, suited crops, benefit score
articles.csv	62	The searchable documents
symptoms.csv	76	Symptom keywords fed to Aho-Corasick
synonyms.csv	396	Local and Hindi names mapped to our word
smalltalk.csv	9	Greetings and everyday replies

9. Running it

Double click FarmAssist.bat. It opens its own terminal window, compiles the project and starts Sophie inside it. Windows Terminal is used when it is installed, because it provides UTF-8 and 24 bit colour with no setup; a plain console window is used otherwise, widened first so the page fits. Either way the console is switched to UTF-8 so the box drawing renders, and the original code page is restored on exit.

Or by hand, from the project folder:

```
javac -encoding UTF-8 -d out (every .java under src)
java -Dstdout.encoding=UTF-8 -cp out ui.ConsoleChat data
```

A JDK is the only requirement; the project was built and tested on JDK 25.

10. Demonstration script

Typed in this order, this covers all eight algorithms and every screen.

Type this	What it demonstrates
algo demo	All eight algorithms on inputs small enough to check by hand
hi then what can you do	The everyday conversation layer
paddy	Synonym rewrite, and the crop record being searchable
my tomatoe has yelow leaves and brown spots	Edit Distance, then Aho-Corasick, then KMP, then QuickSort
suggest fertilizer for tomato under 3000	0/1 Knapsack
match fertilizers for rice cotton banana	Bipartite Matching
which crop for low rainfall	The crop planner
compare rice and wheat	Two crops side by side
how to grow rice	Rabin-Karp, KMP, QuickSort, then Suffix Array + LCP
stem borer	Pest profile and control
trace off	Hides the algorithm trace for a clean screen
list crops list pests list diseases	The database behind it